

Master Thesis Proposal

Detecting Social Group Mentions in Parliamentary Speeches with LLMs

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1 Motivation

Politicians try to win votes by strategically targeting the social groups that they believe make up their voter base. In the field of electoral competition, research has focused mainly on policy appeals (Dolezal et al., 2014). These are explicit statements used by political parties to promote certain policies that they are confident their voters will support. Appeals to social groups, on the other hand, function on a more symbolic level and are aimed at people’s social identification of being part of a larger collective group of people (Tajfel, 1974). When making political decisions, people rely either consciously or unconsciously on their social identity (Kinder and Kam, 2010). Political actors take advantage of this fact by deliberately appealing to these social identities to signal that they represent the interests of the group in order to win their votes. Research has shown that these references can actually alter people’s evaluation of candidates (Robison et al., 2021).

Policy appeals and group appeals are not mutually exclusive and can be used in combination rather than in competition to each other. A political actor might try to appeal to a social group by talking about a particular policy that his or her party supports and that will benefit the group’s interests. However, as previous studies have shown, group appeals are also important in their own right, as they are increasingly being used by parties in election campaigns (Thau, 2019).

In the past, the detection of group appeals in political texts was mainly a task of manual content analysis (Thau, 2019; Dolinsky, 2023). Researchers worked their way through party manifestos and classified each sentence by hand, which is a very time-consuming process and naturally limits the research to a rather small corpus. Dictionary-based approaches have been applied to facilitate this process (Riethmüller and Franzmann, 2024) while still having a major drawback. There are many ways to address a social group. It is simply not possible to capture all possible linguistic nuances in a dictionary. Either the dictionary is too narrow and omits several alternative formulations of the same appeal, or it is too broad and introduces a large amount of noise (false positives) into the set of identified mentions.

For this reason, recent advances in the field of Large Language Models (LLMs) can be used to automate the process of recognizing mentions of social groups while capturing the different nuances with which these mentions are made. In addition, it is necessary to classify whether the political actor has tried to associate their party with the group in question or whether they have tried to signal their disapproval in order to convince other people who also

express a dislike of the mentioned group. This can be addressed with an aspect-based sentiment analysis, that can either be adopted by an encoder-based or generative LLM (Hoang et al., 2019). Some scholars classify merely the presence of a social group mention (Licht and Sczepanski, 2024). Ideally however, one does also detect which exact group has been mentioned. Previous research predefined several group categories in order to train an LLM to categorize already detected group mentions into one or more of these classes (Horne et al., 2025). However, I argue that it is difficult to know all these classes a priori and propose an inductive approach to classify the group mentions. After passing the references through an embedding model, one could apply a clustering method such as DBSCAN to classify them into meaningful groups in an unsupervised manner. I will describe my proposed classification pipeline in more detail in the Data and Methods section of this proposal.

So far, group appeals have mainly been analyzed for manifesto data. This makes sense, as it is to be expected that the parties communicate most directly in these texts which policies they advocate and which social groups they support. How individual politicians use group appeals in their own communication with the electorate has so far been neglected. I therefore propose to examine how individual elected MPs use group appeals to communicate with their party’s electorate, but also with their local constituencies in which they were voted in. The extent to which MPs are allowed to pursue their own agenda or whether they are forced to closely follow the party’s general line of communication depends to a large extent on the country’s institutional framework (Proksch and Slapin, 2012). Examining the communications of British MPs would be an interesting case, as the UK’s first-past-the-post electoral system provides MPs with significant incentives to stick to their local constituencies, as they are the only ones who can get the MP re-elected. (Cain et al., 1987). I propose to analyze British legislators’ strategic appeals to social groups within parliamentary speeches. These speeches are well documented and available as there exist several data sources such as the ParlSpeech V2 dataset (Rauh and Schwalbach, 2020) which provides comprehensive coverage of parliamentary speeches for a wide range of countries over for a long time frame. A longitudinal analysis of the concept would also make it possible to examine whether the political strategy of mentioning social groups in parliamentary speeches is as pronounced as researchers claim it is in party manifestos.

To summarize, I propose to develop a classification pipeline that automatically detects appeals to social groups and classifies the expressed sentiment towards the mentioned group. Based on the extracted text embeddings, I would apply clustering to categorize the passages into meaningful classes, which can then be manually labeled through qualitative inspection. My specific contribution would be to further develop the automated extraction of social group appeals and apply this method to a so far rather neglected topic in this field, namely the individual communication of politicians in parliament over time.

2 Literature Review

In the past, appeals to social groups were mainly examined on the basis of manual content analysis. Thau (2019) hand-coded British election manifestos of the Labour Party and the Conservative Party from 1964 to 2015. He concludes that group appeals have always been a strategic tool used by British parties to win votes and that the use of these tools has increased over time. Another study by the same author concludes that class differences in vote choice respond directly to the extent to which the British Labour Party resorted to group appeals to

the working class (Thau, 2021). Huber (2022) has extended the concept of group appeals in general by establishing an appropriate definition of when specific appeals might be beneficial to a party. She finds support for her theoretical expectations by examining manifesto data from Austria. Dolinsky (2023) recognizes the need for a longitudinal cross-country dataset on group appeals and develops such by analyzing manifestos and print campaigns of parties from Israel and the Netherlands between 1977 and 2015. Stuckelberger and Tresch (2024) consider other data sources such as press releases and television advertisements in their analysis of group appeals around elections in Germany, Switzerland and the Netherlands. They explain the use of strategic appeals, by the general support of the parties by social group, and use survey data to check their assumptions.

While there are relatively few studies that apply a dictionary-based approach for the detection of group appeals (Riethmüller and Franzmann, 2024), a growing number of scientists are discovering the potential of LLMs to study the concept. Licht and Szczepanski (2024) develop a supervised classification approach by using BERT models to detect social group mentions at the token level. However, they neither classify the mentions' sentiment nor the specific addressed group. These limitations have tried to be solved in a recent working paper by Horne et al. (2025) who create a classification pipeline in which they classify manifesto data at the sentence level according to whether the sentence contains a group mention, its sentiment as well as the respective group that is being addressed. Each group mention can fall into one or more of a set of predefined group classes the scholars predefined. While these works all rely on fine-tuning encoder LLMs to perform classification, further studies demonstrate the possibility of having a generative LLM perform this task by few-shot prompting without any fine-tuning (Riethmüller et al., 2024; Jalali Farahani et al., 2024).

3 Theoretical Framework

For an accurate classification of social group mentions it is essential to clearly define what a social group is. I would mostly rely on the foundational definition by Huber and Dolinsky (2023) who understand a social group as a "cluster of people with common ascriptive or acquired socio-demographic characteristics, including ethnicity, race, gender, age, sex, education, occupation, income, class, nationality, family status, religion, and region". I would extend this rather rigid definition by the inclusion of groups that share more general attributes like common values and life experiences (e.g. "honest people", political beliefs) Licht and Szczepanski (2024). This definition still excludes any institutional groupings such as parties, firms or (trade) unions. However, I would follow Huber and Dolinsky (2023) by classifying an explicit mention of the people within these institutions as a social group. Secondly, a social group mention does not need to be positively appealing to the respective group. By speaking negatively about people in question the speaker might try to implicitly connect to a completely different group. This distinction between either positive or negative group mentions motivates the sentiment analysis.

The reason why politicians make group-based appeals to the electorate is because they see this as a successful strategy to persuade citizens to vote for them in the upcoming elections. Social identity theory states that people see themselves as members of different social groups (Tajfel, 1974; Tajfel et al., 1979). Moreover, people use these social group memberships to improve their self-image and self-esteem. This happens by associating positive emotions and values with one's own social group, while negative emotions are associated with other

groups. The result is in-group and out-group thinking that does not necessarily lead to conflict but has the potential to do so when these groups are in competition with each other. In applying this psychological concept to politics, scholars find that people use their social group membership and their respective in-group and out-group distinctions when thinking about political issues and making political decisions (Conover, 1988; Kinder and Kam, 2010). People do so because group references can be a cognitive shortcut for humans when making political decisions that are oftentimes complicated and would require a careful evaluation of all candidates (Thau, 2019). Instead, voters use these shortcuts to find out which politicians claim to represent the social group to which they themselves feel they belong.

Analyzing the use of group appeals by politicians at the individual level also requires an understanding of the factors that influence MPs' incentive to seek personal votes rather than simply to convey the party's main lines. Most crucial to this incentive is the country's electoral system. When citizens elect candidates directly, as in the British first-past-the-post system, the incentive for individual politicians to speak up and represent the interests of the local constituency increases (Proksch and Slapin, 2012, 2014). Furthermore, electoral vulnerability serves as another factor that might explain why particular legislators are particularly engaged in vote-seeking and targeting the local electorate (André et al., 2015). Electoral vulnerability is usually measured as the percentage gap between the election result of the winner of a parliamentary seat and the best opposing candidate. So if MPs have won their seat by a narrow margin, they have a particular incentive to appeal to their local voters to increase their chances of re-election. However, if MPs represent a district that can be considered a safe seat, they may not be so keen to win personal votes.

4 Hypotheses

Based on the theoretical framework outlined above, I have developed the following working hypotheses. They aim to examine how group appeals are used at the aggregate party level and at the level of the individual politician.

H1: The total proportion of group mentions used by legislators increases as an election approaches.

The first hypothesis targets the aggregate level and intends to test the general assumption that group appeals are used as an election strategy for parties to win votes. If this is the case, the proportion of these references should increase before an election.

H2: The effect of H1 is more pronounced among legislators who won their seats by a narrow margin in the last election.

The need to win votes is likely to be particularly acute for MPs whose seats are at risk in the next election, meaning that they are more likely to appeal to group identities than MPs who are not worried about being reelected.

H3: The type of group mentions used by legislators depends on their party's voter base.

While the first two hypotheses focused on the general use of group appeals without considering their specific content, this hypothesis formulates expectations regarding the particular nature

of the groups addressed. Parties should direct positive appeals to their voter base and negative appeals to groups that they expect their voter base to reject.

H4: The type of group mentions used by individual legislators depends on the party's voter base as well as the legislators' local constituency.

Here, I would like to analyze once again the individual level at which legislators use group references. These should be influenced by the core ideology of their party and their voter base, as well as by their local constituency, since it is ultimately these citizens who are responsible for electing the politician in question. However, it is unlikely that politicians will attempt to appeal to social groups that do not belong to their party's electorate. Instead, I would expect to see a strengthening effect in local constituencies with a particularly high proportion of the relevant voters.

Overall, I have to conclude that the hypotheses still need to be reconsidered theoretically and are by no means definitive. In order to examine *H3* and *H4*, I would have to determine which voter groups the parties are composed of, which of the social groups I have identified this can be transferred to, and which other social groups are rejected by these people.

5 Data and Method

I intend to analyze parliamentary speeches made in the House of Commons. The Parlspeech V2 Dataset ([Rauh and Schwalbach, 2020](#)) comprises all speeches that were given in this parliament between 1988 and 2019. By webscraping the official transcription website of the British parliament it would also be possible to collect the speeches for the period after 2019. My proposed method for classifying appeals to social groups would be as follows. From the entire dataset, I would randomly sample sentences to manually label whether and where the sentence contains a mention of a social group. I would also annotate whether the particular mention was made with a positive or negative sentiment. These should form the basis for a training set to fine-tune a generative LLM for identifying mentions of social groups at the token level. To further increase the training set, a generative LLM could be tasked with augmenting the sentences for which one or more group references were found in order to obtain a greater variety of sentences with which to fine-tune the model. It would be necessary to retain a certain portion of the original manually annotated sentences to create a validation and a test set to test the performance of the model against a ground truth.

If the performance of this first model is sufficient, I would apply it to the entire corpus to classify each individual sentence according to whether it contains one or more group mentions and where exactly these are located. I would then perform an aspect-based sentiment analysis for each recognized group mention to classify whether the reference was actually a positive appeal or whether the person was signaling opposition to the respective group. For this task, I could fine-tune another generative LLM using the hand-coded annotations from the first sample. This model could then be queried for each individual extracted group mention. The goal is to end up with a data frame containing all sentences that contain a group appeal, together with the sentiment expressed towards the group. If a sentence contains multiple group references, each of these references would result in a separate row in the data frame. Finally, to classify the detected group mentions into meaningful group categories I propose to apply an unsupervised approach. One could use an embedding model to get a representation of each group mention in a high-dimensional vector space. These embeddings could then be

passed as input to a clustering algorithm such as DBSCAN (which would have the advantage of not having to set the number of clusters manually). The result of the clustering would then have to be checked manually in order to assign a meaningful group name to each cluster. This inductive approach would have the advantage that no assumptions would have to be made about which group categories might be present in the data set. This would also allow group categories to be identified that had not been thought of before.

6 Limitations and Open Questions

I still have a few questions that I need to clarify before starting the project, and I can already see some limitations I would like to address.

First of all, my original motivation was primarily to develop a solid classification pipeline for appeals to social groups that works with a different source of political texts than previous studies. Since scholars have only analyzed manifesto data so far, I also thought it would be insightful to examine how these group references are used on an individual level. However, my hypotheses still require some theoretical consideration. At present, I am not defining which parties have which voter bases, which social groups these bases consist of, or who they are directed against. Since this study is primarily intended to make a methodological contribution, I think it makes sense to narrow the scope somewhat and focus on one or two social groups where there is little dispute about which British parties claim to represent these people. This could be, for example, the connection between the working class and the Labour Party in the United Kingdom. On the other hand, a longitudinal analysis would need take into account how such group alignments have changed over time. In all cases, such claims would need to be further substantiated by literature and possibly by survey results from the country itself.

I see another limitation in the interpretation of the clustering results. While I would have distinguished between individual group mentions, it is entirely possible that a single mention could refer to several groups at once, making its position in the embedding space difficult to interpret. An example of this would be the fictional group reference “young working mothers.” Such a mention would affect three groups at once (young people, working class, parents). The alternative to the unsupervised approach I have presented would be to define the existing group categories in advance and assign and classify each group reference directly to one or more of these groups.

One open question I still need to clarify is which LLMs I actually want to use for classification and sentiment analysis. I haven’t decided yet and need to consider whether it makes sense to pay a certain price for a proprietary model or whether the performance of open-source models is sufficient. Finally, I suggested fine-tuning the generative LLMs for the task. An alternative would be to use in-context learning so that the LLM can make the decision without specific training.

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